

Finite element modeling of subcutaneous implantable defibrillator electrodes in an adult torso

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BACKGROUND Total subcutaneous implantable subcutaneous defibrillators are in development, but optimal electrode configurations are not known.

OBJECTIVE We used image-based finite element models (FEM) to predict the myocardial electric field generated during defibrillation shocks (pseudo-DFT) in a wide variety of reported and innovative subcutaneous electrode positions to determine factors affecting optimal lead positions for subcutaneous implantable cardioverter-defibrillators (S-ICD).

METHODS An image-based FEM of an adult man was used to predict pseudo-DFTs across a wide range of technically feasible S-ICD electrode placements. Generator location, lead location, length, geometry and orientation, and spatial relation of electrodes to ventricular mass were systematically varied. Best electrode configurations were determined, and spatial factors contributing to low pseudo-DFTs were identified using regression and general linear models.

RESULTS A total of 122 single-electrode/array configurations and 28 dual-electrode configurations were simulated. Pseudo-DFTs for single-electrode orientations ranged from 0.60 to 16.0 (mean 2.65 ± 2.48) times that predicted for the base case, an anterior-posterior

configuration recently tested clinically. A total of 32 of 150 tested configurations (21%) had pseudo-DFT ratios ≤ 1 , indicating the possibility of multiple novel, efficient, and clinically relevant orientations. Favorable alignment of lead-generator vector with ventricular myocardium and increased lead length were the most important factors correlated with pseudo-DFT, accounting for 70% of the predicted variation ($R^2 = 0.70$, each factor $P < .05$) in a combined general linear model in which parameter estimates were calculated for each factor.

CONCLUSION Further exploration of novel and efficient electrode configurations may be of value in the development of the S-ICD technologies and implant procedure. FEM modeling suggests that the choice of configurations that maximize shock vector alignment with the center of myocardial mass and use of longer leads is more likely to result in lower DFT.

KEYWORDS ICD; Defibrillation; Modeling

ABBREVIATIONS DFT = defibrillation threshold; FEM = finite element models; ICD = implantable cardioverter-defibrillator; S-ICD = subcutaneous implantable cardioverter-defibrillator.

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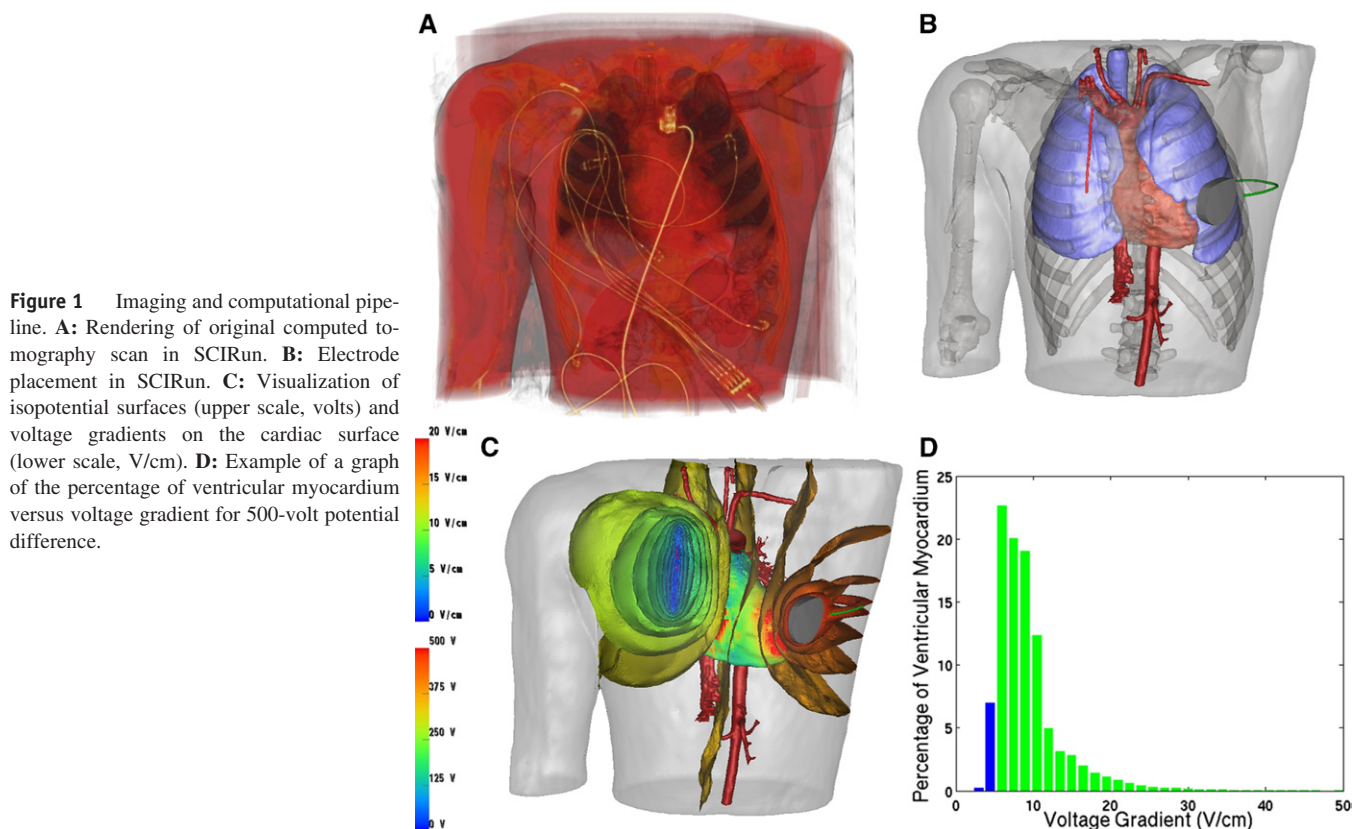
Introduction

Implantable cardioverter-defibrillators (ICDs) have become the standard of care for patients at risk of fatal cardiac arrhythmias, and indications for their use continue to expand.^{1–3} In current practice, ICD systems typically use transvenous ICD leads in a standard superior vena cava–right ventricle coil orientation. Extracardiac ICDs using subcutaneous defibrillation leads (S-ICD) are currently in development.^{4–6} This concept has generated interest because it might provide alternative ICD implant strategies allowing more widespread application

of ICD therapy for the primary prevention of sudden death.⁷ Avoidance of transvenous systems would also obviate vascular and cardiac complications associated with lead placement and extraction.^{8,9}

Reports of extracardiac implantation using standard ICD technologies adapted for use in patients with small size and congenital heart disease have shown that a defibrillation approach based on subcutaneous implants can be clinically effective.^{10,11} Nontransvenous defibrillation efficacy has also been explored in human adult studies using several different shock vectors.¹² Preliminary studies of these shock vectors, which are contingent on the anatomy of electrode implantation in relation to the thorax, have demonstrated that defibrillation can be performed with achievable shock energies.⁵ However, these vectors have largely been selected by trial and error, and no systematic efforts have been made to optimize their efficiency.

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We have previously described a finite element computer modeling program that can be used to explore the electric fields generated by electrodes placed in arbitrary locations and orientations in the human torso as a metric for estimating the relative defibrillation efficacies of subcutaneous ICD electrode orientations.^{13,14} We now describe the utilization of this system to explore a wide variety of published and novel orientations to further guide S-ICD development. The hypothesis for this study was that with the rapid exploration of electrode positions made possible by our system, we would be able to identify critical factors in S-ICD design and to optimize the efficacy of electrode orientations.

Methods

Finite element modeling methodology

The methods used for image segmentation, model construction, and prediction of defibrillation efficacy have been previously described¹³ and are briefly summarized. With internal review board approval, a torso model was created from a 1.25-mm-slice computed tomography scan of a young adult man with normal cardiothoracic anatomy, obtained from a trauma database. The torso was segmented into tissue compartments using the 3D Slicer (Surgical Planning Laboratory, Brigham and Women's Hospital, Boston, Massachusetts; <http://www.slicer.org>) and imported into SCIRun 4.1 for electrode placement and solution of the bioelectric field problem (Scientific Computing and Imaging Institute, University of Utah, Salt Lake City, Utah; <http://software.sci.utah.edu/>). Custom software developed

allowed interactive placement of electrode models into the segmented anatomy with clinically realistic degree of precision (Figure 1).

The segmented volume and electrode models were rendered as a hexahedral mesh for finite element modeling. Tissue-specific conductivity values derived from literature were then projected onto the computational mesh.^{15,16} Electrodes were assigned a constant potential over their surface, and the finite element model solved using standard techniques, resulting in the assignment of a predicted voltage gradient to each element of the finite element models (FEM).^{13,17–19} This model is static and calculates the predicted electrical field as an unvarying, steady-state phenomenon; the entire defibrillation waveform and associated effects of tissue capacitance were not modeled.

Relative defibrillation thresholds (pseudo-DFTs) are defined as the primary outcome measure for this study, and are a measure of the interaction of the electrical field generated by an ICD shock with the ventricular myocardium. The pseudo-DFT is calculated for each electrode configuration as the shock energy $E = \frac{1}{2} CV^2$, where C is the estimated capacitance of a typical pulse generator (130 μ F), and V is the electrode voltage necessary to achieve likely defibrillation. This was determined by application of the critical mass hypothesis, which proposes that a defibrillation shock will be successful if it produces a threshold voltage gradient over a large fraction of the myocardial mass. The criterion used in this study was a voltage gradient of 5 V/cm generated over 95% of the ventricular myocardial elements of the

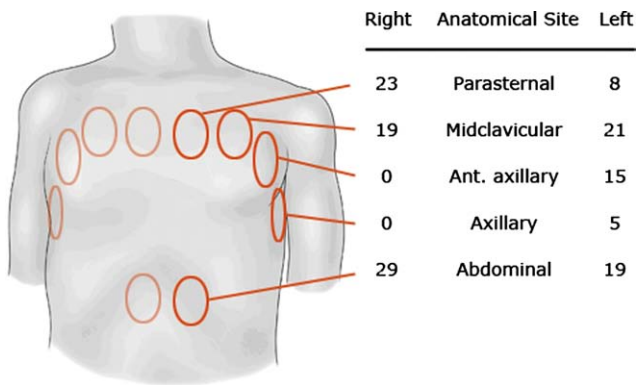


Figure 2 Left-sided implantable cardioverter-defibrillators can positions corresponding to named parameters, with the number of configurations tested in both left- and right-sided versions of each position.

FEM, parameter values that have been accepted as a reasonable predictor of successful defibrillation.^{18–21} Although the pseudo-DFT does not include an active model of cardiac defibrillation and may not accurately estimate the clinical DFT, this metric can be used as a yardstick by which the intrathoracic field strength over the myocardium can be compared given differing electrode configurations, which is a controllable factor affecting defibrillation.²²

Modeling of clinically relevant electrode configurations

Using the torso model, the location, laterality, and orientation of the electrodes (generator and subcutaneous leads), and the lead number, polarity, and length were systematically varied. Electrode positions were referenced to standard surgical anatomy. Examples of locations used are presented in [Figure 2](#).

Specific electrode configurations tested, although not exhausting all possible anatomical permutations, were chosen by a broad, developmental approach, inclusive of several variations of all promising configurations. Ease or plausibility of surgical implantation was not used for a priori exclusion of novel configurations. All previously reported electrode configurations were modeled after images presented in the literature. Models containing 2 leads were studied in all polarity permutations, including a nonactive ICD can, except in cases in which the electrodes were

Table 1 Parameters used to describe electrode configurations tested

Parameter	Value range
Electrode level (generator or lead)	L2–T4
Electrode laterality (generator or lead)	Right/left
Generator site	Axillary, anterior axillary, midclavicular, parasternal, abdominal
Lead site	Parasternal, anterior, anterolateral, lateral, posterior, posterolateral
Lead orientation	Horizontal/vertical/L-shaped
Lead number	1/2/3
Lead length	5 cm/10 cm/15 cm/20 cm/25 cm
Minimum interelectrode distance	Continuous (Figure 4 , dimension B)
Minimum distance electrode to epicardial surface (generator or lead)	Continuous (Figure 4 , dimensions C and D)
Minimum distance primary interelectrode vector to ventricular center of mass	Continuous (Figure 4 , dimension A)

specified to function as an array; in these cases, the lowest pseudo-DFT was assigned to that orientation.

For assessment of the relative efficacy of each configuration, the predicted pseudo-DFT was normalized to a base case: the pseudo-DFT determined for a documented subcutaneous configuration that has been clinically tested in patients.⁵ This anteroposterior model used an active can in a left low parasternal position and a 25-cm coil electrode tunneled around the back of the left thorax between the 6th and 10th intercostal space, with 80% of patients having a clinically determined DFT between 20 and 30 J in this orientation ([Figure 3](#), left).

Statistical analysis

Each electrode configuration was characterized with respect to the descriptors presented in [Table 1](#). Continuous relations of the electrode and can to the heart were measured as shown in [Figure 4](#).

The predicted pseudo-DFT was taken as the outcome variable; this variable was log-transformed to meet assump-

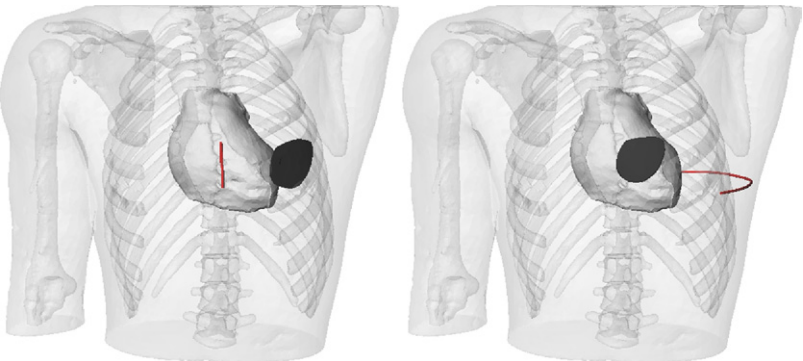


Figure 3 Starting orientation for Grace et al⁶ (left) and Lieberman et al⁵ (right) in the modeling environment based on the literature. All modeled orientations were normalized to the predicted DFT for the orientation on the right (Lieberman et al⁵). DFT = defibrillation threshold.

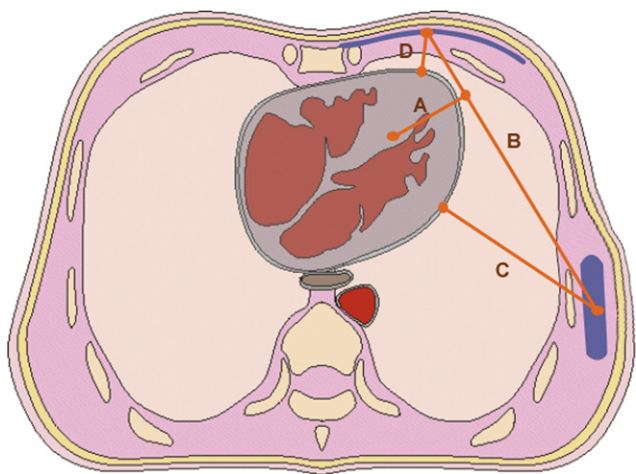


Figure 4 Diagram of distances measured for each electrode and generator configuration. **Metric A:** Alignment of thoracic field with myocardium, measured as the distance of the center of mass of the heart from the line between generator and lead. **Metric B:** Minimum distance between generator and lead. **Metric C:** Minimum distance between generator and surface of heart. **Metric D:** Minimum distance between lead and surface of heart.

tions of linearity and to permit statistical analysis. Univariate analyses of parameters describing system anatomy were performed using linear regression for continuous variables and Wilcoxon rank sum and Kruskal-Wallis tests with Bonferroni correction for dichotomous and categorical variables (SAS9.1.3, SAS Institute Inc Cary, North Carolina, 2002 to 2005). The Spearman correlation was chosen for independence of the result from parameterization or assumptions of linearity. Parameters associated with predicted DFT with a value of $P < .05$ were determined by stepwise linear regression to determine the most important parameters, and both the total R^2 and individual parameter partial R^2 values were calculated. These were then integrated into the general linear model and estimates for each parameter as well as the various levels within that variable.

Results

Association of anatomical parameters with predicted DFT

The approach outlined above resulted in assessment of 230 different configuration/polarity combinations (122 single electrode, 108 dual electrode). A total of 122 distinct single-electrode/array configurations and 28 dual-electrode configurations were simulated. Pseudo-DFT ratios (normalized to base case) ranged from 0.60 to 16.04 (mean 2.65 ± 2.48). A total of 32 of 150 tested electrode configurations (21%) had pseudo-DFT ratios ≤ 1.00 , suggesting that multiple novel, efficient, and clinically relevant orientations may be exist other than those previously reported.

Univariate modeling results suggested that a variety of anatomical factors affecting the geometry of system configuration influenced pseudo-DFT. Placement of the generator in the parasternal position was more efficient than more lateral and remote positions (midclavicular, anterior axil-

lary, abdominal). Anterolateral and posterior electrode positions were better than parasternal, and anterolateral better than anterior. Right-sided generators were more efficient than left-sided generators, whereas the converse was true for electrode laterality.

Multivariate modeling using a linear regression model showed that variability in the predicted pseudo-DFT was highly attributable to alignment of electrode generator shock vector with ventricular myocardium (metric A squared) and electrode length together ($R^2 = 0.70, P < .0001$). A general linear model was then applied to incorporate categorical variables into a combined model with a cutoff for significance for each individual term of $P < .05$. The general linear model incorporating the identified factors simultaneously showed that favorable alignment of shock vector with ventricular myocardium (metric A squared), increased lead length, can horizontal position, contralateral lead-generator position, and distance of can from the heart (metric C) resulted in a combined model with $R^2 0.825 (P < .0001)$, and parameter estimates and associated P values for each term were calculated (Table 2). Metric A squared and electrode length remained the 2 largest contributing factors to the total R^2 of the model and remained robust through all data exploration; the imputed effects of variation of these factors on pseudo-DFT are shown in Figure 5.

Investigation of previously described orientations

System proposed by Lieberman et al⁵

This S-ICD approach proposes a low, medial pectoral position of an active generator and a 25-cm posterolateral electrode extending around the back of the left thorax between the 6th and 10th intercostal space, extending the tip as close to the spine as possible (Figure 3, left). This constituted the base case we used for normalization of predicted pseudo-DFT (i.e., pseudo-DFT ratio = 1). Varying electrode length from 10 to 25 cm, the vertical and lateral positions revealed that the original 25-cm electrode, positioned posteriorly at either the T8 or T10 level, remained the

Table 2 General linear model inclusive of all factors

Parameter	Parameter estimate	P level
Metric A (squared) Field alignment with heart	176	<.0001
Lead length	−4.09	<.0001
Generator position	*	<.0001
Metric C Proximity of generator to heart	6.37	.0014
Opposite	*	<.0001

$P < .05$, total model R squared 0.83, $P < .0001$. To achieve linearity, log transformation of pseudo-DFT was performed. $\ln(\text{pseudo-DFT}) = 2.59 + a(\text{Metric A})^2 + b(\text{lead length}) + c(\text{generator position}) + d(\text{metric C}) + e(\text{opposite})$. All distances in meters for calculation of parameter estimates. DFT = defibrillation threshold.

*Parameter estimates for categorical variables implied static differences in the levels of the variables, but none resulted in large changes in pseudo-DFT.

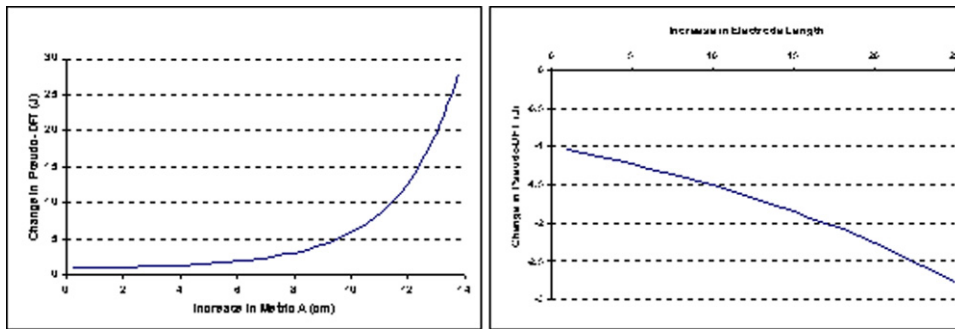


Figure 5 Predicted effects of 2 major parameters on predicted DFT based on the general linear model. **Left:** Effect of increase in Metric A (worsening of alignment of shock vector with ventricular myocardial center of mass) to increase predicted DFT. **Right:** Effect of increase in electrode length to decrease predicted DFT by smaller increment. DFT = defibrillation threshold.

most efficient, with pseudo-DFT ratios ranging from 1 to 1.73.

System proposed by Grace et al⁶

This S-ICD system has been depicted as an active generator positioned in the anterolateral axillary line in the 6th intercostal space, paired with a parasternal electrode approximately 3 cm left of the sternal midline. Electrode length has not been specified. We modeled this configuration as shown in Figure 3 (right) and varied the length and position of the parasternal electrode. Holding the can position constant, we placed a 5-cm electrode in the left parasternal region and varied its vertical position from T4 to T8, which resulted in pseudo-DFT ratios of 7.3 to 9.8 compared to the base case, left parasternal electrode in the left parasternal region and varying its vertical position from T4 to T8, resulting in pseudo-DFT ratios from 7.3 to 9.8, compared with the base case. These values were reduced to 2.3 to 3.1 when the electrode position was right parasternal. Lengthening the right parasternal electrode to 10 cm in this right parasternal position resulted in a further decrease in pseudo-DFT ratios

to 1.6 to 2.0, and use of a 15-cm electrode (extending from T4 to curve around the xiphoid at T10) resulted in a pseudo-DFT ratio of 1.22.

Best electrode orientations

Single-electrode orientations

Examples of the most efficient single-electrode situations modeled are shown in Figure 6, top panels. Similar to the general vectors that were described by Lai et al,²³ orientations with a generator located on right upper chest can and a long anterior electrode position on the anterolateral left lower chest electrode performed well. Left-sided thoracic generator placement was efficient when the long electrode coil was extended posteriorly around the chest wall (as in Lieberman et al,⁵ and illustrated in panel).

Two-electrode orientations

Exemplary, efficient 2-electrode orientations are shown in Figure 6, bottom panels. In these configurations, the pseudo-DFT ratio represents the optimal of permutations of cathode and anode assignment.

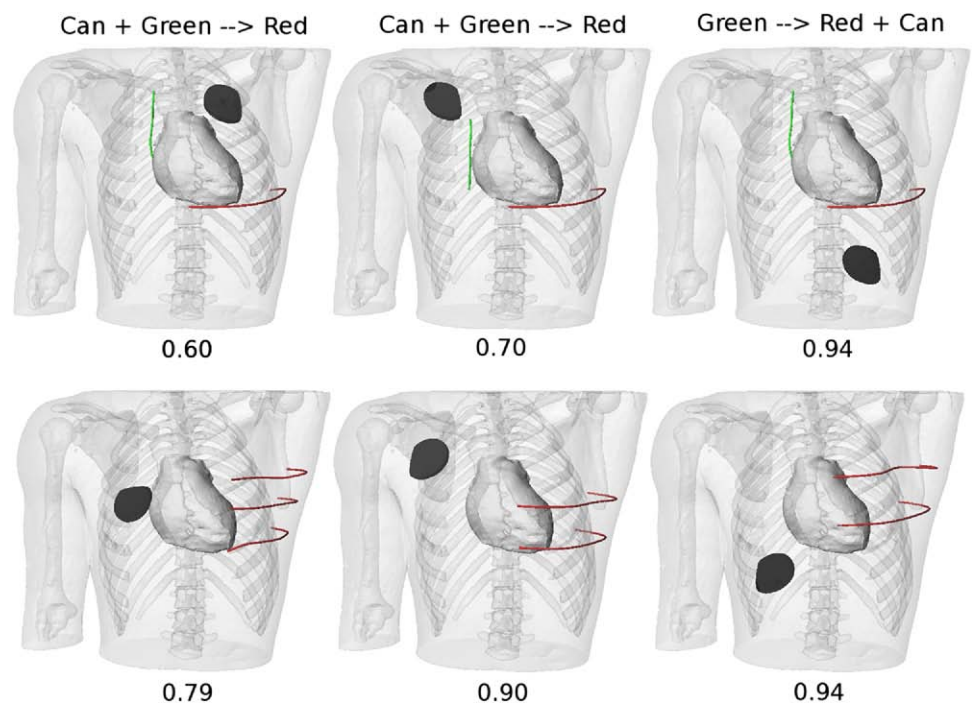


Figure 6 **Top panels:** Examples of 2-electrode configurations with predicted DFT ratios. **Bottom panels:** Examples of electrode array configurations with predicted DFT ratios. DFT = defibrillation threshold.

Electrode arrays

The majority of efficient adaptations of electrode arrays involved placement of the generator on the right chest and electrode coils around the lateral aspect of the left lower chest. Use of a right abdominal can with left lateral electrodes was also efficient (pseudo-DFT ratio 0.97).

Discussion

In 2004, over 100,000 ICDs were implanted in the U.S. alone, a 60% increase from 1997. This number may be expected to increase given broadening indications for such devices.^{3,24–26} There has been interest in the development of S-ICDs, in part because availability of simpler devices and implantation techniques that reduce or eliminate the need for fluoroscopy and lead placement skills will further expand access to ICD technology. Defibrillation technology of this type also will be of value to children and to congenital heart populations, as current transvenous approaches to defibrillation are often not well suited to their anatomies.

Data from early efficacy studies indicate that S-ICDs will be required to deliver higher-energy shocks than transvenous systems. Although such energies are achievable with existing technology, this requirement poses a challenge to S-ICD designs, as trade-offs will exist among the defibrillation energy allocated per shock, size of the device, and predicted battery life. Development of S-ICD configurations that maximize defibrillation efficiency thus will have favorable electronic consequences for their design.

In this study, a wide variety of surgically feasible S-ICD electrode configurations were tested in a finite element model of the human torso to investigate the effects of anatomical variation on the electrical field induced in the ventricular myocardium. Two previously published S-ICD configurations were modeled, with 1 validated configuration⁵ selected as the base case for normalization of the predicted DFTs of all tested configurations. The significant findings of the current study are as follows. First, a wide variety of conceivable electrode orientations, some of them quite unusual and not previously reported, are predicted to be as effective or more effective than the base case (pseudo-DFT ratio ≤ 1). Second, although some of these alternatives represent simple modifications of the previously proposed system, many involve changes in lead design and implant technique that are substantial; in particular, the contralateral placement of generator and lead.

Finally, multivariable analysis of the predicted DFT against the parameters used to specify the anatomical locations of the electrodes revealed that 2 primary design principles—placement of the electrodes to align the interelectrode shock vector as closely as possible to the center of mass of the ventricular myocardium, and use of longer electrode coil lengths—account for the majority of variability in the predicted DFT. The relative positions of the generator, the lead(s), and the ventricular myocardium accounted for nearly half of the predicted variability in the pseudo-DFT. This reflects the intuitive observation that electrodes should be positioned to place the heart as nearly

between them as possible. Manipulation of electrode length contributed almost 25% of the variability, with decreases in pseudo-DFTs predicted with extension of coil length from 5 to 10 cm and longer. Neither of these factors has previously been quantified for subcutaneous electrode placement and may prove useful in determining optimal orientations. Notably, although electrode arrays were often identified as useful in many efficient configurations, the use of an array was generally not necessarily more efficient than a single electrode of equal length similarly positioned. This finding implies both for S-ICD design and for current subcutaneous arrays used to augment transvenous systems with unacceptably high DFTs that a simple, single-electrode system is likely to offer as much benefit as an array, which is more difficult to implant and may be more prone to failure.

After characterizing the factors most important in determining the pseudo-DFT ratio, we looked at 2 specific orientations suggested in the literature. The configuration created to emulate that proposed by Grace et al⁶ in their original report was relatively inefficient, but it was possible to significantly decrease the predicted pseudo-DFT by lengthening the subcutaneous lead and moving it to the right parasternal area. The configuration proposed by Lieberman et al⁵ used an anterior generator and a 25-cm coil electrode extending posteriorly to the spine. This orientation has been clinically tested and constituted our base case for normalization of our predicted pseudo-DFTs. Several single-electrode and array configurations with DFT ratios ≤ 1.0 were identified, generally involving contralateral placement of electrodes (generator on right chest wall, subcutaneous leads on left chest wall).

Widespread clinical use of an S-ICD will require further proof of defibrillation efficacy, adequate ventricular sensing and detection algorithms, and system reliability, and also must be easy to implant. Ease of implantation and the possible effects of lead anatomy on longevity/failure rate are critical counterbalancing factors of importance to the mechanical design of such systems and the development of implant procedures, and these may offset, for example, any beneficial electrical effects of placing an electrode in locations contralateral to the generator. Ideally, S-ICDs would be implanted over as limited a surgical field as possible, and in proximity to vascular structures used for transvenous systems, to facilitate transition to transvenous approach if necessary.

The limitations of FEM as a predictor of DFTs have been discussed previously.^{13,22} Our model assumes homogenous conductivity in tissue compartments, linearity of electrical response, and negligible tissue capacitance effects. It is important to reiterate that although we use the term pseudo-DFT for the purpose of comparing various electrode configurations, that estimate reflects the geometric interaction of electrical field and the ventricular myocardium and does not incorporate patient- or tissue-specific factors that will influence the true, clinically determined DFT. Ultimately, a more sophisticated approach will integrate models such as

ours, which represent the anatomically inhomogeneous volume conductor of the thorax within which implantable devices generate electric fields interacting with the heart, and bidomain simulations of the active defibrillation process in the ventricular myocardium.^{22,27} However, to date this more sophisticated time-dependent modeling incorporates only cardiac factors, and does not take into account the volume conductor factors of the surrounding tissues, which are likely even more important to a subcutaneous system than traditional transvenous or epicardial orientations due to the current having to travel through noncardiac tissues, including poorly conducting lung, before reaching the heart. With extension to whole-torso models and validation, such complex models may be able to provide more accurate and unbiased estimates of clinical defibrillation thresholds. Further modeling in female, obese, and pediatric patient models is needed to determine the effects of sex, size, and body habitus. Lastly, our statistical analysis and modeling were based on a dataset created with optimization and best orientations in mind and is skewed by a prior knowledge of good orientations. A more distributed dataset including more positions across a broader range of positions would be necessary to make stronger associations with more certainty.

Conclusion

We have developed an interactive computational and visualization tool and used it to compare the relative efficiency of proposed electrode orientations for subcutaneous ICD implantation, as well as to suggest novel orientations that might also be of value. These studies suggest several new implantation strategies and the utility of the development of new electrodes, as well as highlighting the potential value of *in silico* modeling in device development and procedure planning.

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